William Glazer-Cavanagh

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Professional Summary

Applied ML Engineer | **Bridging Research & Product**

Software Engineer and ML Scientist with 3 years of experience bridging research and production systems. I have built optimized language model inference servers, deployed robust ML pipelines, and delivered reusable research tooling and workflows.

Skills

Modeling: PyTorch | JAX | Transformers | Object Detection | RAG | Fine-tuning | Learning to Rank
Infrastructure: Docker | Modal (Serverless) | Dask | Parquet/Zarr | Distributed Systems
Core Stack: Python | Bash | ML System Design | A/B Testing | GPU Profiling | Git | OOP & Functional Programming | Unit Testing

Work Experience

AbCellera, Montreal/Vancouver | *Machine Learning Scientist II* — **05/2023 - Current**

I built ranking and filtering models for protein engineering workflows, where every data point costs thousands. My work spanned ranking systems, low data regimes, language models, noisy biological assays, and distributed GPU inference.

- **Risk Prediction Product:** Led the full lifecycle of a sequence filtering tool, successfully deploying a backend product that reduced downstream experimental failure rates by 30%.
- **Language Model Inference Server:** Increased throughput by 100x by enabling horizontal scaling, vectorizing pre-processing and eliminating cold start latency.
- **Model Calibration:** Calibrated foundation model likelihoods to ensure confidence scores aligned with empirical failure rates.
- **Evaluation & Ablation Design:** Created ablation studies with curated datasets and rigorously A/B tested against baselines.
- **Model Testing:** Curated golden datasets for automated testing pipelines, preventing regressionsz on different machines.

I built high-throughput computer vision models for microscopy, processing 100,000+ images daily. I optimized models to the physical instrument limits across noisy, complex assay formats.

- **High-Throughput Vision Architecture:** Designed and deployed a lightweight deep learning architecture (MobileNet backbone + detection head) that maximized imaging throughput from 400 to 1200 images/s, hitting the hardware's physical limits, while increasing model accuracy.
- **Data Pipelines & Active Learning:** Built a semi-supervised pipeline to label complex multi-class assays, achieving an 80% preference rate from human experts in blind A/B tests. Doubled performance on edge cases (0.40 to 0.95 mAP) by building internal developer tools to fine-tune models on hard examples.
- **Infrastructure & Optimization:** Eliminated training bottlenecks for 1TB-scale microscopy data by engineering a sharded format (PyArrow/Parquet/Zarr) and leveraging Dask for parallel processing to fully saturate GPU utilization.
- **MLOps:** Ensured reproducible experiments by managing configurations with Hydra, tracking experiments and models with CometML, and versioning data with DVC.

Croesus, Montreal | *Research Intern* — **05/2021 - 08/2021**

I reduced LLM hallucinations in generative AI stock price summaries by leveraging financial news data.

- **Applied Research & RAG:** Adapted open-source research to build RAG workflows, integrating financial Knowledge Graphs into BERT-based models.
- **Evaluation & Benchmarking:** Evaluated state-of-the-art Information Retrieval (IR) papers against established baselines to identify and integrate top-performing architectures.

Education

MSc. Machine Learning, Professional Masters | *Graduation Year 2023* — **University of Montreal (MILA)**

- **Benchmarked JAX vs. PyTorch:** Re-implemented reinforcement learning algorithms (TD3/VPG) using JAX (Haiku/Optax) to evaluate JIT compilation benefits. Accelerated training by 5x using XLA compilation compared to standard PyTorch (JIT vs. Eager).

BEng. Software Engineering, Artificial Intelligence Profile | *Graduation Year 2022* — **Polytechnique Montreal**

- **Published Research:** Co-authored *Change Taxonomy for ML Pipelines* (Empirical Software Engineering, 2023).
- **Community Leadership:** Co-founded PolyAI, a student initiative for Applied AI.